

Your submitted solution is only **valid** if it contains the following paragraphs at the top of the program. Please *fill* in *your name* and *Neptun code*.

```
/*<NAME>, <NEPTUN>
```

```
This solution was prepared and submitted by the student stated above for the assignment of the Programming course. I declare that this solution is my own work. I have not copied or used third party solutions. I have not passed my solution to my classmates, neither made it public.
```

```
Students' regulation of Eötvös Loránd University (ELTE Regulations Vol. II. 74/C.§) states that as long as a student presents another student's work - or at least the significant part of it - as his/her own performance, it will count as a disciplinary fault. The most serious consequence of a disciplinary fault can be dismissal of the student from the University.*/
```

When submitting on TMS upload the program with the output enabled that only prints the result of the specific subtask e.g.: if you are solving a) do not include the outputs for tasks b), c), d)!

Stock Portfolio Analysis

A financial analyst tracks the weekly closing prices for several stocks over a period of time. The input is structured as a list of records, where each record contains the stock ticker followed by a series of weekly closing prices. Multiple records may exist for the same stock.

Input

- The first line of input contains n ($1 \leq n \leq 10$), the total number of stock records tracked, and w ($1 \leq w \leq 100$), the number of weekly closing prices provided for each record. This line also contains a Threshold Price ($10 \leq p \leq 500$).
- Each of the next N lines contains:
 - The Stock Ticker symbol (e.g., TSLA).
 - w floating-point numbers separated by spaces representing the weekly closing prices for that stock record.

Write a program that solves the following tasks:

Tasks:

- Find the first weekly price record that is exactly equal to the Threshold Price p . Print the name of the stock and the index of the matching week. If no price meets this criterion, output -1.
- Find and print the index and the ticker symbol of stock record where the difference between its lowest and its highest recorded weekly closing prices is the biggest. If there are several equal matches, choose the stock of the first appearance.
- Output the total number of unique stocks tracked. Each subsequent line should display the stock's ticker symbol followed by the sum of all its weekly closing prices (aggregated from all records belonging

to that stock), separated by a space. Maintain the order of their first appearance in the input.

- d. Print the number of unique stocks with *at least one* price greater than p across all their associated price records. Then, list each of these stock ticker symbols once, in the order of their first appearance.

Output

Each task's output should be separated by new lines. Do *NOT* print the text in red; the red color is used here solely to illustrate and distinguish each subtask's output for clarity.

Test Case 1

Input

```
4 5 50
TSLA 80 45 90 55 10
GOOG 51 52 50 50 53
TSLA 100 10 110 15 20
AAPL 60 70 40 50 50
```

Output

```
Task a):
GOOG 3

Task b):
3 TSLA

Task c):
3
TSLA 535
GOOG 256
AAPL 270

Task d):
3
TSLA
GOOG
AAPL
```

Test Case 2

Input

```
5 6 75
AMZN 90 80 60 77 55 120
MSFT 40 50 74 65 88 70
NFLX 95 85 66 72 99 59
NVDA 110 60 58 89 76 69
META 55 73 66 60 71 74
```

Output

```
Task a):
-1
Task b):
1 AMZN
Task c):
5
AMZN 482
MSFT 387
NFLX 476
NVDA 462
META 399
Task d):
4
AMZN
MSFT
NFLX
NVDA
```

Evaluation based on 10 input tester files:

Points: $10 \cdot (2+2+3+3) = 10 \cdot 10 = 100$ points